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FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2025

THERMO-CHEMICAL HEAT TRANSFER

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.

ii. Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Discuss the causes and consequences of incomplete combustion. Use an appropriate example to substantiate your answer. How is the possibility of incomplete combustion addressed in modern combustors?	**1**	**3**	**10**
	b.	Methyl alcohol (CH_3OH) burns with excess air at an air-fuel ratio (mass) of 8. Determine the equivalence ratio, the mole fraction of CO_2 in the product mixture assuming complete combustion.	**1**	**2**	**10**
2.	**a.**	Discuss the significance of determining the adiabatic flame temperature. What factors cause the actual flame temperature in practical combustion systems to differ from the adiabatic flame temperature? Elaborate on these factors.	**2**	**3**	**10**
	b.	Discuss chemical equilibrium and the conditions for equilibrium.	**2**	**2**	**10**
3.	**a.**	Differentiate between lower and upper heating values of a fuel. Calculate the heating value (HHV and LHV) for methane (CH_4) when burned in stoichiometric proportions at STP. Given Data: Enthalpy of formation (@298K, in kJ/kmol): CH_4: -74831, H_2O (g) :-241845, CO_2: -393546 Enthalpy of vaporization of water (kJ/kmol): 44010	**3**	**2**	**10**
	b.	Discuss the mechanism of vapour release from liquid and gaseous fuels during their combustion with examples.	**3**	**1**	**10**
		OR			
4.	**a.**	Explain the concept of flammability limit and discuss the factors affecting flammability limits.	**3**	**1**	**10**
	b.	In different practical applications, pre mixed and diffusion flames are employed. Discuss in detail and differentiate between pre-mixed flames and diffusion flames, highlighting their salient features.	**3**	**2**	**10**

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5.	**a.**	Discuss Stefan Boltzman's law in radiation. Use the same to determine the black body emissive power of a suspended metallic ball with 25-cm-diameter, having a temperature of 500° C. What will be the total amount of radiation energy (in Joules) emitted by the entire ball in 5 minutes?	**4**	**2**	**10**
	b.	Consider a wall of a building made of brick, with dimensions of 4 m in height, 5 m in width, and 0.3 m in thickness. Calculate the rate of heat leakage if the temperature inside the room is maintained at 22°C, while the outside temperature is initially 0°C. If the outside temperature drops further to -10°C, determine the thickness of insulation required on the interior side of the wall to maintain the indoor temperature at 22°C. You can use an insulating board of thermal conductivity 0.02 W/mK on the interior side of the wall. Assume that the interior and exterior heat transfer coefficients remain unchanged at 10 W/m^2K and 30 kW/m^2K respectively. Thermal conductivity of brick may be taken as 0.9 W/mK. Comment on the effect of the thermal conductivity of the insulation board on the thickness choice.	**4**	**2**	**10**
		OR			
6.	**a.**	Discuss the importance of thermal conductivity and thermal diffusivity in selecting materials for thermal insulation.	**4**	**3**	**10**
	b.	Discuss Newton's law of cooling and convective heat transfer coefficient. Explain the physical significance of Nusselt number in determining the heat transfer coefficient in convection.	**4**	**3**	**10**
7.	**a.**	Establishing a safe distance from fire is crucial in fire safety. Discuss how these safe distance calculations can be made.	**5**	**3**	**10**
	b.	Consider a pool fire with a burn rate of 100 g/s and pan size of 1.5mx 1.5m. The calorific value of the fuel may be taken as 45 MJ/kg. Assume the radiant fraction as 0.3. Estimate the radiant heat flux at floor levels 3,5, and 10m away from the flame. Use Point Source Model.	**5**	**3**	**10**
		OR			
8.	**a.**	Compare Point Source Model and Solid Flame Model.	**5**	**3**	**10**
	b.	A crib fire has an average temperature of 1000°C. Find the heat flux at a target situated at a 3 m distance, if the view factor is 0.01. Discuss the assumptions you will follow to arrive at the solution.	**5**	**3**	**10**